MIL-M-80141C
30 April 1987
SUPERSEDING
MIL-M-80141B
5 September 1974

MILITARY SPECIFICATION

METALLIZING OUTFITS, POWDER-GUNS AND ACCESSORIES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of metallizing outfits which apply coatings in the form of molten spray to metallic or non-metallic surfaces. The molten spray is developed from material in powder form which is heated by a flame produced by oxygen and a fuel gas.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

ZZ-H-461 - Hose and Hose Assembly, Rubber, Gas (Acetylene-Hydrogen, Air and Oxygen).

MILITARY

MIL-V-13877 (AL) - Valve, Regulating, Acetylene and Oxygen

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3433
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MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

MIL-P-83348 - Powders, Plasma Spray.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

2.1.2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR, Chapter XVII, Part 1910 - Occupational Safety and Health Standards.

(Copies of specifications, standards, and other Government publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AEROSPACE MATERIAL SPECIFICATIONS (AMS)

AMS 4026 - Aluminum Alloy Sheet and Plate.

AMS 5504 - Steel Sheet, Strip, Plate, Corrosion and Moderate Heat Resistant.

(Application for copies should be addressed to Customer Service Department, Publications Group, Society of Automative Engineers, 400 Commonwealth Drive, Warrendale, PA 15086-7511.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/RMA/CGA IP-7 - Specification for Rubber Welding Hose.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018-3363.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM C 633 - Adhesion or Cohesive Strength of Flame Sprayed Coating.

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187.)

COMPRESSED GAS ASSOCIATION (CGA)

CGA E-1 - Standard Connections for Regulator Outlets, Torches and Fitted Hose for Welding and Cutting Equipment.

CGA E-4 - Standard for Gas Regulators for Welding and Cutting.

(Application for copies should be addressed to the Compressed Gas Association Inc., PO Box 6208, Falls Church, VA 22046-0907.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The metallizing outfit shall be new and one of the manufacturer's current models capable of spraying all types of commercially available powders. The metallizing outfit shall include all components, parts, and features neccessary to meet the performance specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

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3.2.1 Dimensional systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the outfit. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all dimensions, measurement, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, flowmeters , pressure gauges, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the manipulator shall be graduated in the same system.

3.2.2 Reclaimed materials. The outfit may contain reclaimed materials provided such materials will not jeopardize the intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the manipulator.

3.2.3 Energy efficiency. The outfit and its components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be located conveniently to the operator at his normal work station.

3.2.5 Safety and health requirements. All safety devices normally incorporated in this equipment by the manufacturer for the protection of the operator shall be furnished. All parts, components, mechanisms, and assemblies furnished on metallizing outfits, whether or not specifically required herein, shall comply with all of the requirements of 29 CFR, Chapter XVII, Part 1910 that are applicable to the metallizing outfit itself. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.6 Mercury restriction. The outfits shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the outfits.

3.2.8 Environmental protection. The outfits shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the outfits shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

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3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. All oil holes, grease fittings, and filler caps shall be accessible and identified.

3.2.10 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or outfits.

3.3 Construction. The outfits shall be constructed of parts which are new, without defects, and free of repairs.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the outfits. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the outfits shall be painted in accordance with the manufacturer's standard commercial practice and color.

3.3.6 Threads. All threaded parts used on the outfit and its related attachments and accessories shall conform to FED-STD-H28.

3.3.7 Fuel. Unless otherwise specified, the outfits shall be furnished equipped for connection to acetylene fuel gas. When specified (see 6.2.1), the systems shall be furnished for connection to other fuel gases.

3.4 Components. The outfit shall consist of all components and accessories listed in table I. When a complete outfit is not required, only those components and accessories specified shall be furnished (see 6.2.1).

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TABLE I. Equipment and quantity.

Equipment	Qty
Metallizing gun (see 3.4.1)	1
Powder canister and feed assembly (see 3.4.2)	1
Hose set, fuelgas and oxygen (see 3.4.3)	1
Hose air (see 3.4.4)	1
Flowmeter, fuel gas (3.4.5)	1
Flowmeter, oxygen (see 3.4.5)	1
Flowmeter, air or carrier gas (see 3.4.5)	As regd.
Air control unit	1
Regulator, oxygen	1
Regulator, fuel gas	1
Flashback arrestors	2
Carrying case	1

3.4.1 Metallizing guns. The metallizing guns shall be designed to control fuel and oxygen gases, spray powders, and when applicable, air. The powder cutoff control shall be separate from the powder flow rate control. All necessary connections shall be provided for connection of fuel gas, oxygen, powder, and when applicable, air. The guns shall be designed for both machine mounting and hand operation. Each gun shall be furnished with a fixture for machine mounting.

3.4.2 Powder feed assembly. Unless otherwise specified, the outfit shall be furnished with a remote type powder feed assembly. When specified (see 6.2.1), the powder feed assembly shall be attached to the metallizing gun. The assembly shall deliver powder in sufficient quantity to meet the requirements of 3.5.

3.4.3 Fuel gas and oxygen hose and fittings. Fuel gas and oxygen hoses required for operation shall comply with ANSI/RMA/CGA IP-7. The fittings attached to the hoses shall conform to CGA E-1, connection numbers 020 and 021 or 022 and 023 as applicable. All hoses and fittings required to make the system operable shall be provided. The length of the fuel gas and oxygen hose which connects the spray gun to the flowmeters shall be not less than 15 feet in length. The hose which connects the output of the oxygen and fuel gas regulators to the respective flowmeter inputs shall be not less than four feet in length.

3.4.4 Air hose and fittings. Air hoses and fittings required for operation shall comply with ZZ-H-461. All hose and fittings required to make the system operable shall be provided. The length of air hose furnished shall be not less than 15 feet in length.

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3.4.5 Flowmeters. The flowmeters shall be of the direct indicating type. The maximum allowable error of the flow meter shall not exceed ± 3 percent of full scale. The operating range of the flowmeters shall be within 20 percent to 90 percent of the full scale reading. The flowmeters shall be rated for safe operation up to 150 pounds per square inch (psi). Where glass or other brittle material is used in construction of the flowmeters, a transparent non-breakable safety shield shall be furnished to protect operating personnel. Fuel gas flowmeters shall be color-coded red and oxygen flowmeters shall be color-coded green. The flowmeters may be individually or group mounted.

3.4.6 Air filtration and control system. The air filtration and control system shall include a replaceable or cleanable air filter and an air moisture separator. The separator shall have a suitable device for draining off liquid accumulations. The system shall include a two stage pressure regulator and an air pressure gauge to indicate downstream pressure. The minimum regulator range shall be from 4 to 80 pounds per square inch (psi). The system shall deliver a sufficient quantity of air to meet the requirements of 3.5 from an air source which shall not be required to exceed a pressure of 90 psi and a volume of 40 cubic feet per minute.

3.4.6.1 Regulators. The oxygen and fuel gas regulators shall be of the two stage type, suitable for use with the gases utilized. The acetylene and oxygen regulators shall conform to MIL-V-13877(AL) and table II. When fuel gases other than acetylene are specified, the regulators shall conform to CGA Pamplet E-4 (see 6.2.1).

TABLE II. Regulator requirements.

Characteristics	Acetylene	Oxygen
Inlet pressure maximum, psi	300	3000
Inlet pressure gauge, range, psi	0-400	0-4000
Minimum discharge pressure range, psi	0-30	0-80
Discharge pressure gauge, max., psi	60	100
Discharge pressure gauge, dial calibration (psi), not greater than	1 or 2	5
Discharge pressure variation	±2 lb	±2 lb

3.4.7 Flashback arrestors. The outfits shall be furnished with two flashback arrestors, one for the fuel gas line and one for the oxygen line.

3.4.8 Carrying case. When specified (see 6.2.1), a carrying case made of plastic or metal and equipped with a hinged cover shall be provided. Provisions shall be made for securely fastening the contents. The case shall be of sufficient size to hold the gun, air caps, tools, lubricants, nozzles, and instruction manuals.

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3.4.9 Operating pressure requirements.

3.4.9.1 Acetylene pressure. All guns shall meet the requirements of this specification at acetylene pressures not to exceed 15 psi.

3.4.9.2 Oxygen pressure. All guns shall maintain continuous operation as required herein with an oxygen pressure variation of ± 6 psi from the originally adjusted pressure for a neutral flame setting. Readjustment of the gun valves shall not be required to meet this requirement.

3.4.9.3 Air pressure. When compressed air is used, all outfits shall maintain continuous operation as required herein, without readjustment of the gun valve, with a compressed air pressure variation of ± 10 psi from the recommended operating pressure.

3.4.10 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to NFPA 79. Each outfit shall operate from a 120-volt single-phase, 60-Hz circuit.

3.5 Performance. The outfits supplied under this specification shall equal or exceed the delivery rate and deposition efficiency specified in table III. The powders listed in table III shall meet the requirements of MIL-P-83348.

TABLE III. System spray rate and depositon efficiency requirements.

Type and Class from MIL-P-83348	Compositon	Delivery rate lb/hr	Deposition 1/ efficiency
Type I, class 2	D-Aluminum	4	80
Type I, class 2	H-Nickel, Aluminum	6	90
Type I, class 2	V-Nickel-Chromium-Aluminum	6	80

1/ Deposition efficiency - The ratio of sprayed material that bonds to a specimen to the total weight of material that has been sprayed.

3.5.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.6 Standard equipment. Unless otherwise specified, all equipment normally furnished by the manufacturer on his standard commercial outfit shall be furnished. When specified (see 6.2.1), required additional equipment shall be furnished and fully described.

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3.7 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.8 Repair parts. Repair parts shall be furnished as specified (see 6.2.1).

3.9 Marking on plates, charts, and decals. All words on plates, charts, and decals shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. When these methods are not practical, the instructions shall be applied directly onto the item by environmentally protected decalcomania transfer, metal wrap-around tag, stencil, silk screen, or by any other method suitable for the use intended. Markings shall be as permanent as the normal life expectancy of the item to which they are applied and shall be capable of withstanding the environmental test and cleaning procedures specified for the item.

3.9.1 Lubrication chart or plate. A lubrication chart or plate shall be furnished with each outfit. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.9.2 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each outfit. The nameplate shall contain the information listed below. If the outfit is a special model, the model designation shall include the model of the basic standard outfit and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.10 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.11 Workmanship. Workmanship of the outfit and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.5 Quality conformance inspection. A quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), the quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, and the inspection shall be cause for rejection.

4.6 Examination. The outfit shall be visually examined to determine compliance with all requirements of this specification.

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4.7 Tests.

4.7.1 Operational test. The complete outfit shall be connected to the specified sources of fuel gas, oxygen, and when applicable, air and electricity. The outfit shall then be prepared for metallizing in accordance with the manufacturers instructions. The outfit shall be thoroughly checked for gas leaks and, when applicable, air leaks. The outfits shall be tested by lighting and operating for at least one minute without powder feed, followed by five minutes of operation with powder feed, using one of the powders specified in table III. During this period of operation, fuel gas, oxgen, and when applicable air, pressures shall be varied over the range specified in 3.4.9 to verify proper operation of the gun within this pressure range. All controls and adjustments shall be checked for proper function and all gauges and meters observed to determine proper operation. During this test, the molten material shall be sprayed on one test speciman comforming to AMS 4026 (Alumminum) and one speciman conforming to AMS 5504 (steel). The sprayed coating shall have a uniform appearance. Surface defects of the coating shall be limited to small nodules not to exceed 0.045 inch in diameter and shall not exceed 0.025 inch above the surrounding sprayed surface. The coating shall not contain any:

a. Blisters.
b. Cracks.
c. Chips or loosely-adhering particles.
d. Pits exposing the substrata.
e. Any other evidence of malfunction.

Except during lighting, the gun shall be free from backfire and the flame shall be steady and free from fluctuation with and without powder feed.

4.7.2 Performance test. The outfit shall be tested for metallizing capabilities by spraying three test specimens comforming to AMS 4026 and three specimens conforming to AMS 5504. Each aluminum and each steel speciman shall be sprayed with a different powder selected from table III. Preparation of the specimens and the spray technique shall be the responsibility of the manufacturer and as described in their operating manuals. The outfit to be tested shall be prepared as specified in 4.7.1. The sprayed coating shall conform to 4.7.1. Sufficient spraying shall be accomplished to verify that the spray rate for each material is not less than specified in table III.

4.7.2.1 Bond strength test. When specified (see 6.2.1), a bond strength test shall be accomplished in accordance with ASTM C 633. The substrata sprayed, gases used, composition of powders, and test limits shall be as specified by the purchaser (see 6.2.1).

4.7.2.2 Flowmeter test. Fuel gas, oxygen, and air flowmeter, as applicable, shall be tested to determine conformance with 3.4.5. Flowmeters shall be checked at 20, 50, and 90 percent of full scale. A certificate of calibration issued by the manufacturer may be submitted in lieu of this test.

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4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A or level B preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-W-45562.

6. NOTES

6.1 Intended use. The outfits described herein are intended for use in facilities having a requirement for spraying all types of alloy, ceramic, and metal powders to rebuild worn areas or to provide protection against corrosion.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. First article, if required (see 3.1).

c. If outfit design and construction is required in a specific measurement system (US or SI), state required system (3.2.1).

d. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

e. Additional safety and health requirements, if required (see 3.2.5).

f. Painting, if different (see 3.3.5).

g. Operating fuel, if different (see 3.3.7).

h. Outfit components required, if different (see 3.4).

i. Powder feed assembly, if different (see 3.4.2).

j. If regulators are required for fuel gases other than acetylene, specify gases (see 3.4.6.1).

k. Carrying case, if required (see 3.4.8).

l. Electrical system and input power, if different (see 3.4.10).

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m. Additional performance features, if required (see 3.5.1).

n. Standard equipment, if different (see 3.6).

o. Specify and fully describe optional equipment, if required (see 3.7).

p Specify applicable repair parts, if required (see 3.8).

q. Nameplate, if different (see 3.9.2).

r. First article inspection, if different (see 4.4).

s. Quality conformance inspection, if different (see 4.5).

t. Bond strength test, if required (see 4.7.2.1).

u. If preservation, packing, and marking in accordance with MIL-W-45562 is required, specify level of preservation and packing required (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operators manuals, parts lists, wiring diagrams, and acceptance test reports should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Subject term (key word) listing.

Aluminum coating
Coatings
Copper coating
Flame spraying
Metal coating
Metallizing outfits
Metal finishing
Metal protection
Spray coating
Spraying
Surface finishing
Tin coating
Zinc coating

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6.5 Cross-reference of classification changes. The outfit's classification changes from the previous issue of this specification are as follows:

MIL-M-80141	MIL-M-80141A	MIL-M-80141B	MIL-M-80141C
Type I -	Type I	Type I	Deleted
Type II -	Type II	Type II	Deleted
Type III -	Type III	Type III	Deleted
Class 1 -	Class 1	Class 1	Deleted
Class 2 -	Class 2	Class 2	Deleted

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Air Force - 99

Review activities:
Air Force - 71, 84
DLA- GS

User activities
Navy - MC, OS, SH

Preparing activity:
DLA - IP

(Project 3433-0099)

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INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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